

Cell annotation

COURSE ON SCRNA-SEQ DATA ANALYSIS

assigning biological labels such as cell types to clusters or individual cells

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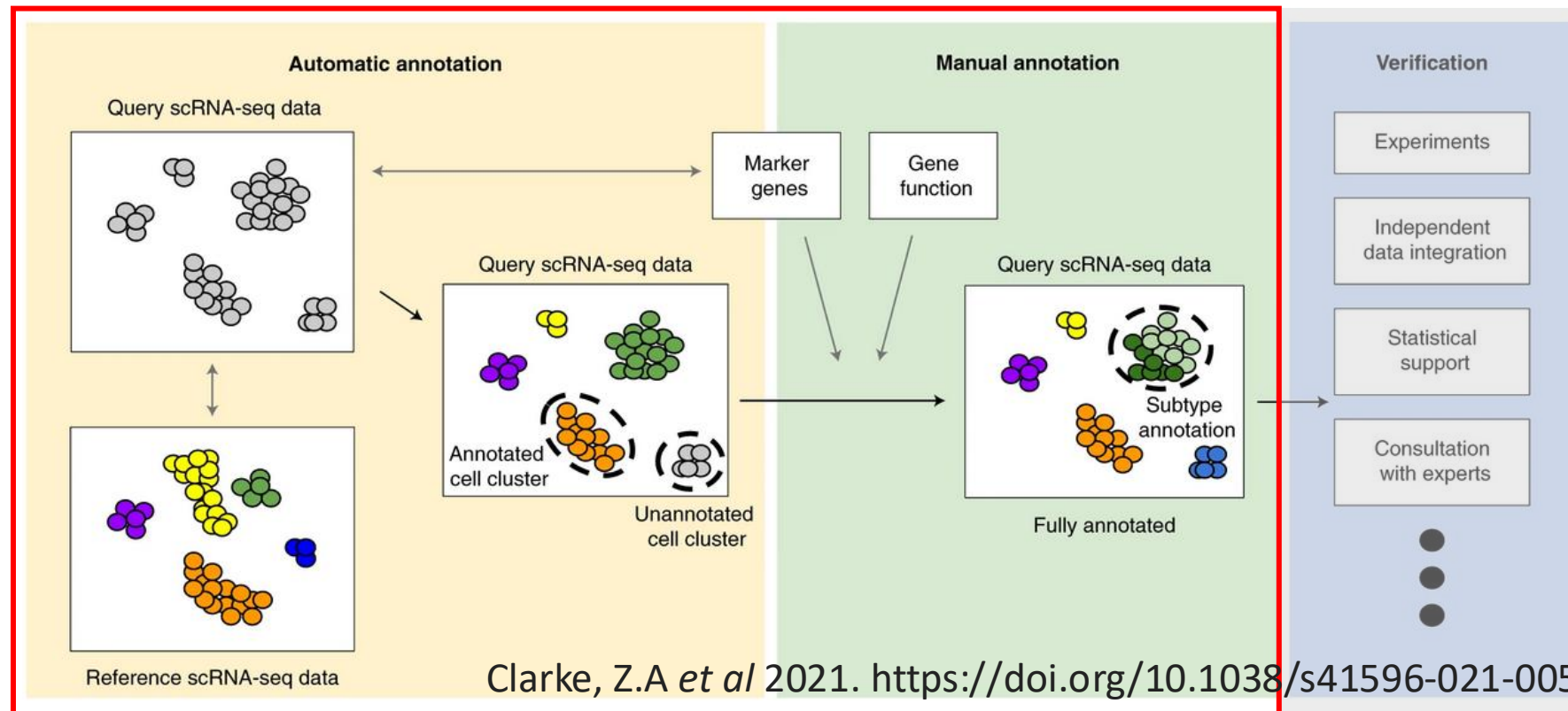
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Learning outcome

- Understand the challenges of cell annotation in scRNA-seq
- Understand different cell annotation approaches

Motivation

exploit prior information I) using marker genes for expected cell types, II) curated gene set associated with specific biological processes III) directly comparison of expression profiles between published reference datasets



Marker-based vs automatic annotation

Marker-based annotation

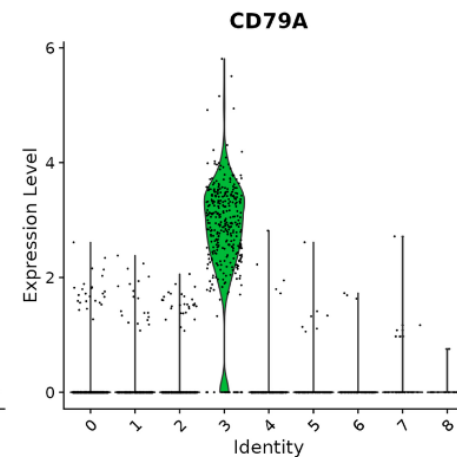
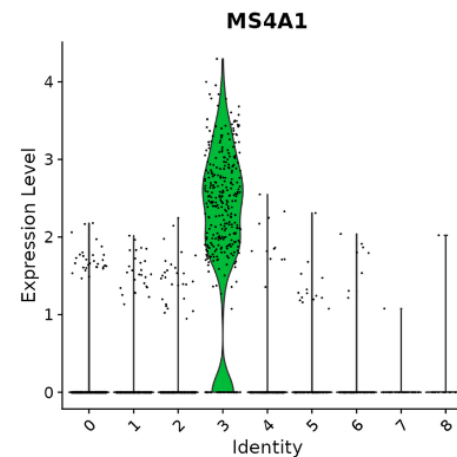
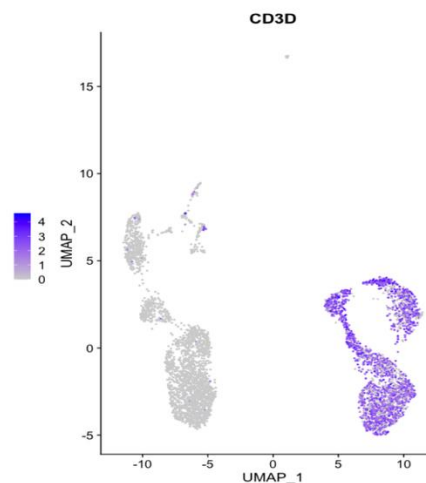
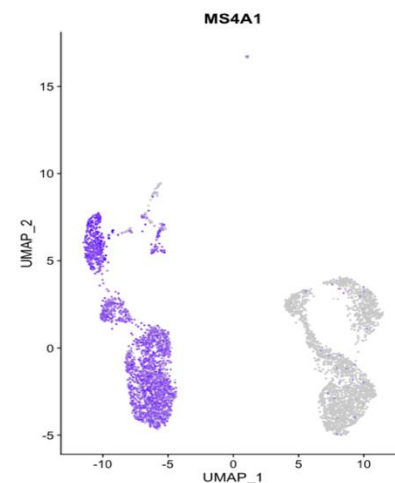
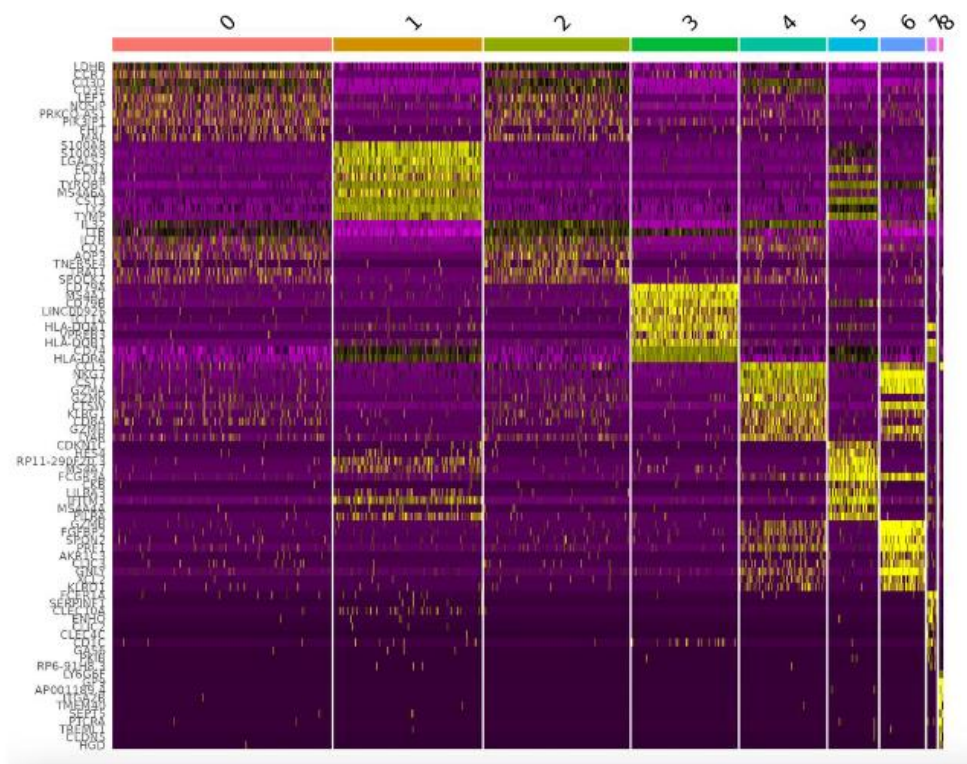
- Time consuming
- Requires expert (or at least prior knowledge)
- Might be subjective

Automatic: requires a reference

- Either use complete cell type-specific mRNA expression profiles based on bulk RNA-seq from FACS-sorted or use of a reference of manually curated cells picked from scRNA-seq data sets
- Can miss cell types if they are not included in the reference

Marker-based annotation

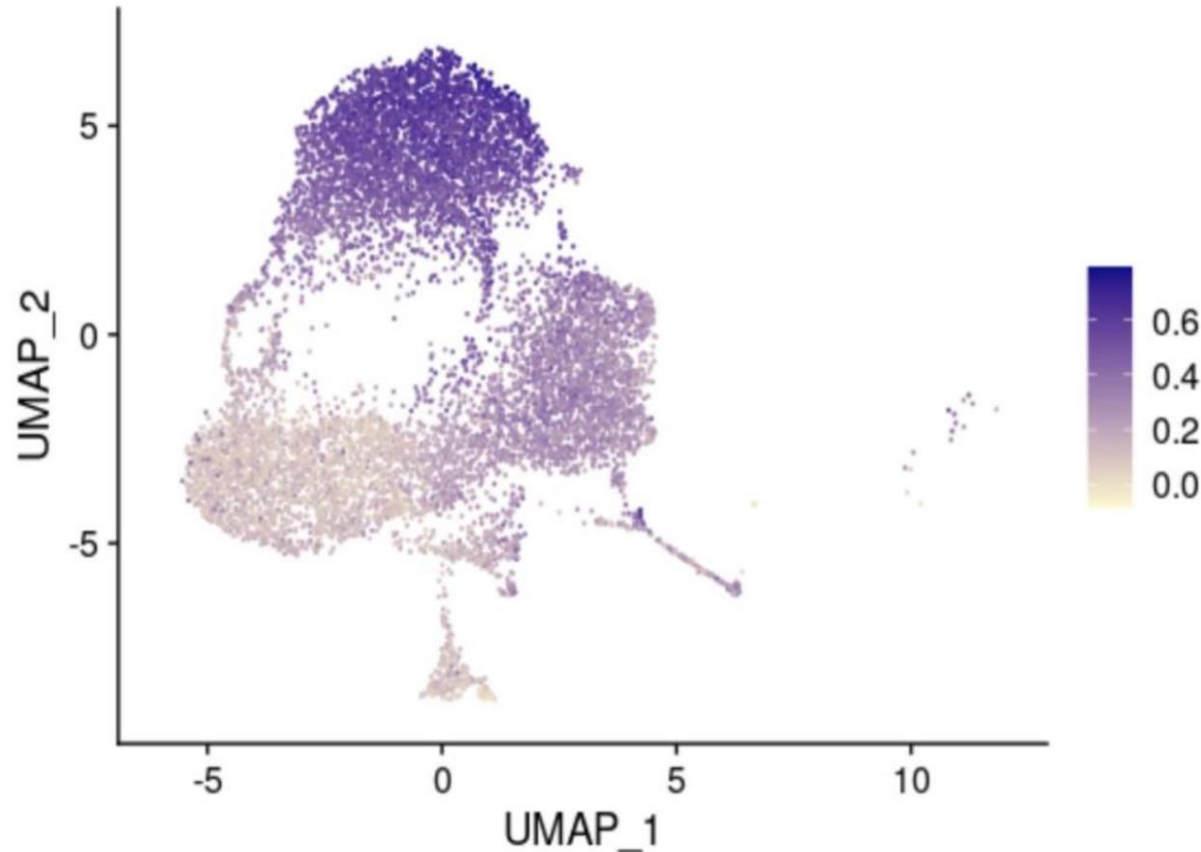
Finding differentially expressed features (cluster biomarkers)



Seurat AddModuleScore

	orig.ident	nCount_RNA	nFeature_RNA	Tcell	Myeloid	NK	Plasma_cell
icHTNA1	Zilionis_immune	7516	2613	0	0.5227121	0	0.00000000
icHNVA2	Zilionis_immune	5684	1981	0	0.5112892	0	0.00000000
icALZN3	Zilionis_immune	4558	1867	0	0.3584502	0	0.07540874
icFWBP4	Zilionis_immune	2915	1308	0	0.1546426	0	0.00000000
							0.00000000
							0.00000000

Score of gene signature

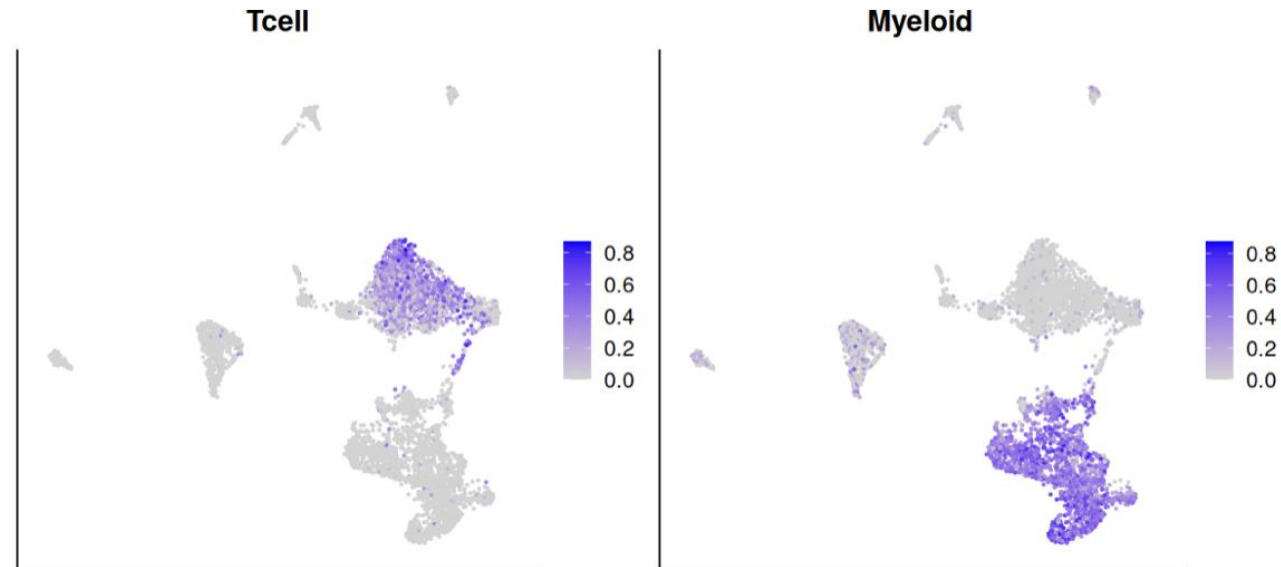


UCell (In Seurat AddModuleScore Ucell)

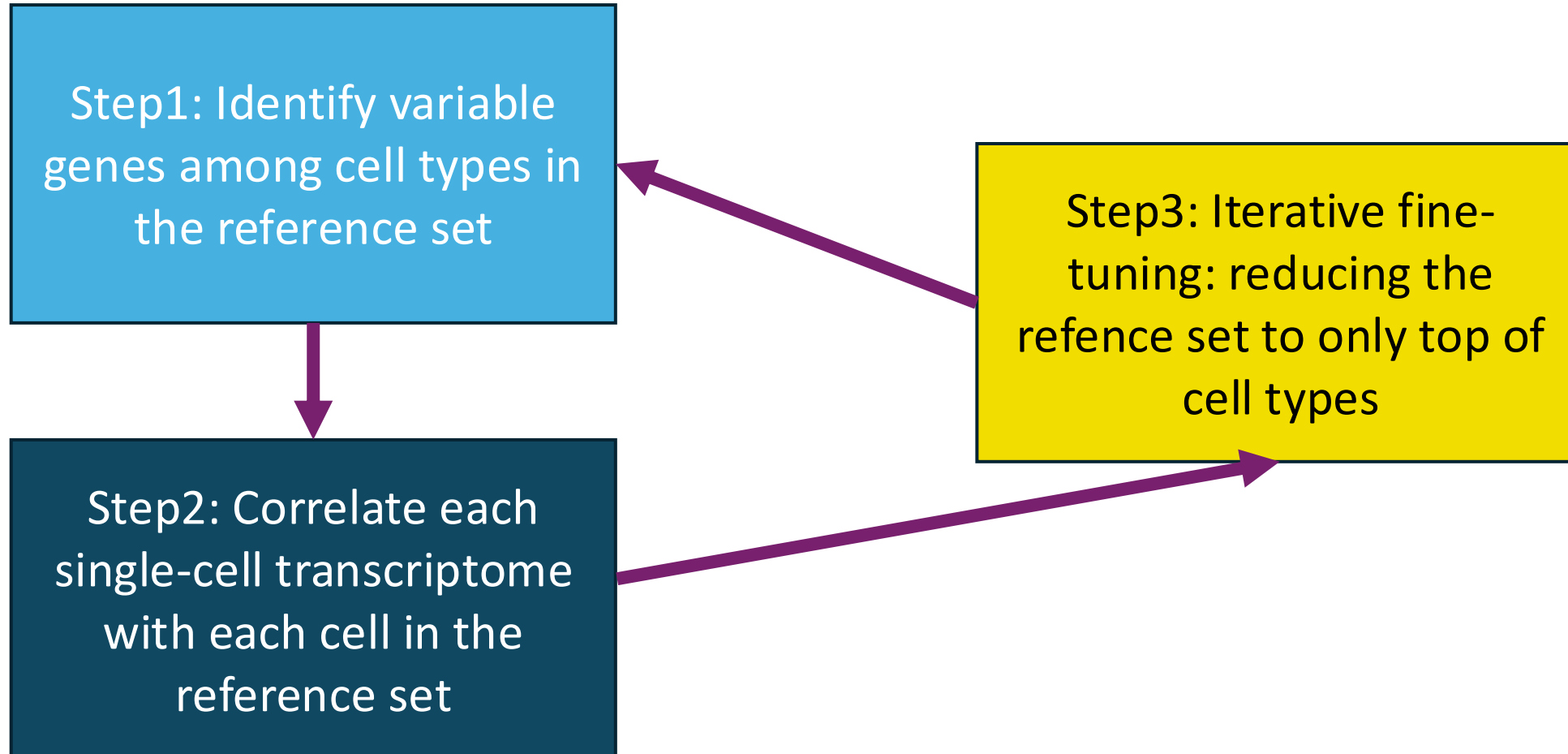
UCell supports positive and negative gene sets within a signature. Simply append + or - signs to the genes to include them in positive and negative sets, respectively. For example:

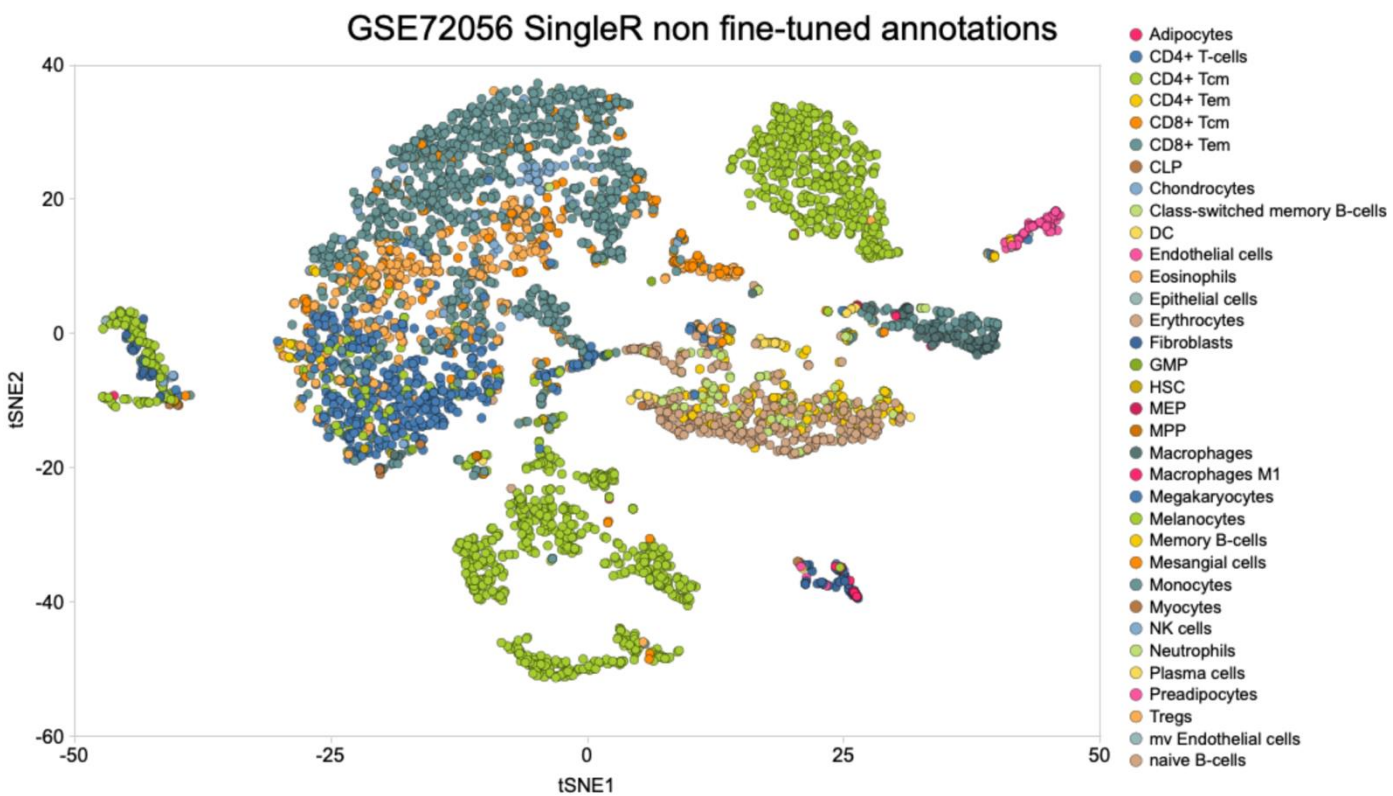
```
signatures <- list(  
  CD8T = c("CD8A+", "CD8B+", "CD4-"),  
  CD4 = c("TRAC+", "CD4+", "CD40LG+", "CD8A-", "CD8B-"),  
  NK = c("KLRD1+", "NCR1+", "NKG7+", "CD3D-", "CD3E-")  
)
```

##		CD8T	CD4	NK
##	bcHTNA1	0.000000	0.14975523	0.000000
##	bcHNVA2	0.000000	0.02503338	0.000000
##	bcALZN3	0.000000	0.00000000	0.000000
##	bcFWBP4	0.000000	0.00000000	0.000000
##	bcBJYE5	0.000000	0.28627058	0.000000
##	bcGSBJ6	0.000000	0.00000000	0.000000
##	bcHQGJ7	0.000000	0.00000000	0.000000
##	bcHKKM8	0.000000	0.21161549	0.000000
##	bcIGQU9	0.000000	0.28649310	0.000000

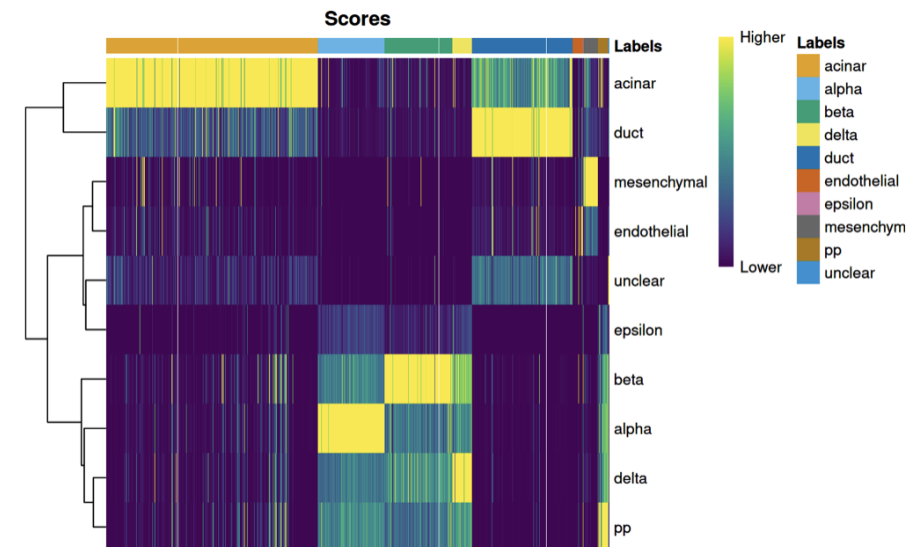


Automatic annotation - SingleR

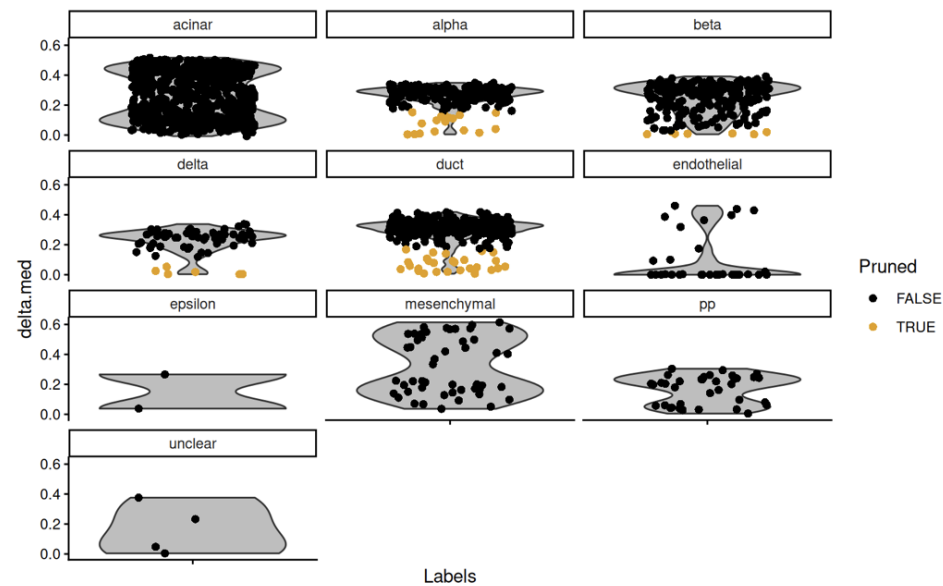




plotScoreHeatmap(pred.grun)

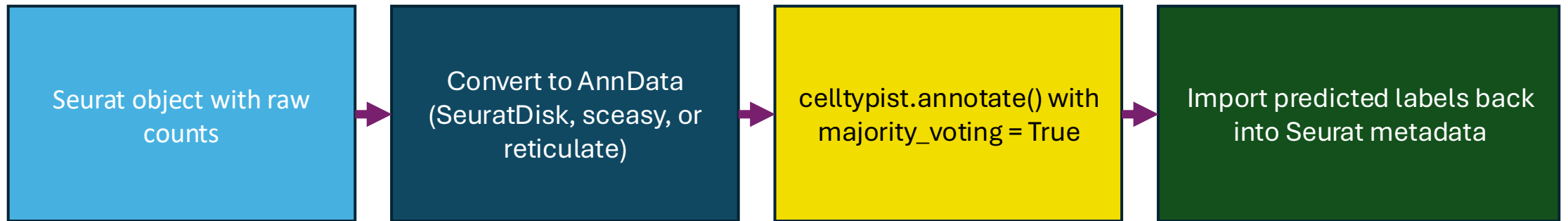


plotDeltaDistribution(pred.grun, ncol = 3)



Automatic annotation – CellTypist

- Assigns cell type labels using pre-trained **logistic regression classifiers**; built-in models currently focus on **immune cell types**, or you can train your own
- **majority_voting** option refines noisy per-cell calls into one consensus label per (over-) cluster
- CellTypist is a **Python** tool (scanpy/AnnData) – from Seurat, a short conversion step is needed first:



Original tool + immune reference: Domínguez Conde, C. et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. Science 376, eabl5197 (2022). Models & docs: celltypist.org, celltypist.readthedocs.io

AI in scRNA-seq annotation

- Traditional cell-type annotation in scRNA-seq depends heavily on expert knowledge or reference datasets that may be incomplete or biased
- AI-based foundation models (*might*) bring automation, scalability, and improved generalization across tissues, species, and experimental platforms.

scGPT

- Pretrained on millions of single cells across diverse tissues
- Learns contextual relationships between genes and cell states
- Performs cell-type annotation, batch correction, data integration, and perturbation prediction
- Works in a zero-shot or few-shot setting
- Reduces need for manual feature engineering

scFoundation

- Unified multimodal embedding space
- Supports cross-modality annotation and integration
- Provides standardized cell-type labels across datasets
- More robust to dataset size, batch effects, and modality noise
- Can transfer knowledge from one omic type to another

Bonus: Database with reference genes/sets

- PanglaoDB <https://panglaodb.se/> (mouse and human), in R: <https://cran.r-project.org/web/packages/rPanglaoDB/index.html>
- CellMarker (mouse and human): <http://bio-bigdata.hrbmu.edu.cn/CellMarker/>
- SingleR <https://github.com/dviraran/SingleR> (Aran et al.), access via cellDex package, e.g. human primary cell atlas (microarrays)
- Human Cell Atlas <https://www.humancellatlas.org> (Regev et al.) single cell RNA seq atlas, also some mouse data
- Single cell portal: https://singlecell.broadinstitute.org/single_cell

- Aran D, Looney AP, Liu L, Wu E, Fong V, Hsu A, Chak S, Naikawadi RP, Wolters PJ, Abate AR, Butte AJ, Bhattacharya M (2019). "Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage." Nat. Immunol., 20, 163-172. doi:10.1038/s41590-018-0276-y.
- Hao Y, Stuart T, Kowalski MH, Choudhary S, Hoffman P, Hartman A, Srivastava A, Molla G, Madad S, Fernandez-Granda C, Satija R (2023). "Dictionary learning for integrative, multimodal and scalable single-cell analysis." Nature Biotechnology. doi:10.1038/s41587-023-01767-y.
- Clarke, Z.A., Andrews, T.S., Atif, J. et al. Tutorial: guidelines for annotating single-cell transcriptomic maps using automated and manual methods. Nat Protoc 16, 2749–2764 (2021). <https://doi.org/10.1038/s41596-021-00534-0>
- Andreatta M, Carmona SJ. UCell: Robust and scalable single-cell gene signature scoring. Comput Struct Biotechnol J. 2021;19:3796-3798. Published 2021 Jun 30. doi:10.1016/j.csbj.2021.06.043
- <https://sib-swiss.github.io/single-cell-r-training/>